

# Hypothesis-Driven Survey Generator

*Turn client research into a fielding-ready questionnaire — with a human in the loop at every step.*

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## What it is

An AI-assisted authoring tool for pharma market researchers that compresses the slowest part of study design — going from a client brief to a structured, validated questionnaire — from days of manual drafting into a guided, reviewable workflow. It does the heavy lifting and asks for sign-off at each stage.

## The problem it solves

Translating research objectives into a well-formed survey is slow, inconsistent, and error-prone: hypotheses get lost, question types don't match what's being measured, routing breaks, and surveys balloon past a fieldable length — and quality depends on whoever happens to draft it.

## How it works — three human-gated phases

- 1 Hypotheses** (Python · Streamlit) — ingests the brief and research docs, asks clarifying questions, and proposes structured hypotheses + a framework. → Researcher approves.
- 2 Survey construction** (Python · multi-agent) — maps hypotheses to sections and drafts questions per section, then runs a **deterministic closed loop**: rule validation → multi-reviewer critique → whole-survey editor → automatic repair that re-validates up to three times until the rules pass. → Researcher approves the outline.
- 3 Rendering** (R · Shiny) — turns the approved outline into a formatted, print-ready Word questionnaire.

## What makes it different

- **A self-correcting engine, not a one-shot generator** — a deterministic rule validator drives revision until the survey is structurally sound; it can't hand you a broken survey.
- **Whole-survey judgment** — a holistic editor optimizes the questionnaire as one instrument (redundancy, consistency, length), not question-by-question.
- **Human-in-the-loop by design** — nothing advances without approval; export is gated until validation is clean (or explicitly overridden).
- **Provider-flexible** — runs on Anthropic, OpenAI, Gemini, or Groq (default Claude Opus 4.8); API keys stay in-session, never written to disk.

## Outputs

Research hypotheses + framework → a clean **Excel survey-outline workbook** (screener, profiling, and body; 17 supported question types with routing, termination, and scales) → a **formatted Word questionnaire**.

**Why it lands.** Consistent, best-practice surveys in minutes instead of days; hypothesis coverage guaranteed; structural quality enforced automatically — with the researcher always in control.